A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-65818135})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1-\sigma_x)^2 I' + \mathrm{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 376 & 102 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,376 + \mathbf{Z}\,(102 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 16454534 = 0$, of discriminant -65818135. Class group invariants $(30 \quad 10 \quad 10)$, class number 3000; the three stored generator ideals (HNF) are $\begin{pmatrix} 167 & 122 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 3236 & 2849 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 1532 & 1449 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1835y^3 + (-4s+2)\,y^2 - 755006y + (4162s - 2081)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 3670*y^8 + 1857213*y^6 + 3034144560*y^4 + 22163904296*y^2 + 2
85029448523735
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2727091}{7515177189376} + \left(\frac{69}{7515177189376}\right) s, \qquad J = \begin{pmatrix} 376 & 102 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 376$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $75609622983685 = 5 \cdot 7 \cdot 2160274942391$:

$$\begin{pmatrix}
75609622983685 & 47283923323083 & 27054639213321 & 34692623375662 & 20080767651362 & 49907583744599 & 45626435912409 & 52729161695551 & 43425964277059 & 44132736917913 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 270 factors with signed exponents of absolute value up to 902920854; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 29$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 3 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1835y^3 + (-4s + 2)\,y^2 - 755006y + (4162s - 2081)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 3670*y^8 + 1857213*y^6 + 3034144560*y^4 + 22163904296*y^2 + 2
85029448523735
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{423739183}{478923089887538176} + (-\frac{134871}{478923089887538176})\,s, \qquad J = \begin{pmatrix} 3436 & 2514 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3436$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $103402703869507 = 7^3 \cdot 41117 \cdot 7331897$:

$$\begin{pmatrix}
14771814838501 & 8966377870163 & 8606216883170 & 8130438787871 & 13343964720712 & 869263280160 & 11400917715298 & 5969108397036 & 6826106309166 & 3300076709857 \\
0 & 7 & 4 & 0 & 2 & 0 & 4 & 3 & 4 & 5 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 276 factors with signed exponents of absolute value up to 3169729789; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 29$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1835y^3 + (-4s + 2)\,y^2 - 755006y + (4162s - 2081)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 3670*y^8 + 1857213*y^6 + 3034144560*y^4 + 22163904296*y^2 + 2
85029448523735
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{9195119}{5023060313370901} + (-\frac{17324}{5023060313370901})\, s, \qquad J = \begin{pmatrix} 1381 & 823 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1381$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $3628600103807 = 7^4 \cdot 1511287007$:

$$\begin{pmatrix} 74053063343 & 7809615821 & 24962335090 & 38114576417 & 68393645561 & 31076094436 & 27497178538 & 64080793374 & 56748121474 & 59310507186 \\ 0 & 7 & 0 & 0 & 1 & 0 & 5 & 2 & 5 & 5 \\ 0 & 0 & 7 & 0 & 1 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 279 factors with signed exponents of absolute value up to 2498889388; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 29$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 4 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 4428y^3 + (52s - 26)\, y^2 - 4028269y + (-7412s + 3706)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 8856*y^8 + 11550646*y^6 + 8818708996*y^4 + 3543006704241*y^2
+ 903974962796860
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2727091}{7515177189376} + \left(\frac{69}{7515177189376}\right) s, \qquad J = \begin{pmatrix} 376 & 102 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 376$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2684026368000 = 2^{12} \cdot 3^8 \cdot 5^3 \cdot 17 \cdot 47$:

$$\begin{pmatrix}
287640 & 284220 & 117360 & 3120 & 101202 & 164874 & 204713 & 234078 & 192594 & 187204 \\
0 & 180 & 84 & 120 & 144 & 48 & 4 & 100 & 97 & 84 \\
0 & 0 & 12 & 0 & 6 & 6 & 3 & 9 & 9 & 0 \\
0 & 0 & 0 & 30 & 12 & 6 & 28 & 20 & 17 & 26 \\
0 & 0 & 0 & 0 & 6 & 0 & 3 & 3 & 5 & 5 \\
0 & 0 & 0 & 0 & 0 & 6 & 2 & 4 & 0 & 5 \\
0 & 0 & 0 & 0 & 0 & 0 & 4 & 3 & 1 & 3 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 207 factors with signed exponents of absolute value up to 472655332384026; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 4428y^3 + (52s - 26)\, y^2 - 4028269y + (-7412s + 3706)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 8856*y^8 + 11550646*y^6 + 8818708996*y^4 + 3543006704241*y^2
+ 903974962796860
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{423739183}{478923089887538176} + (-\frac{134871}{478923089887538176})\, s, \qquad J = \begin{pmatrix} 3436 & 2514 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3436$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $374974030373760 = 2^7 \cdot 3^4 \cdot 5 \cdot 38167 \cdot 189517$:

$$\begin{pmatrix}
1735990881360 & 181384163088 & 1366340857329 & 393394629930 & 1735646724250 & 942187251161 & 1175044230886 & 1588495672502 & 1501797634731 & 866510699935 \\
0 & 12 & 0 & 6 & 11 & 6 & 8 & 8 & 0 & 2 \\
0 & 0 & 3 & 0 & 1 & 1 & 1 & 0 & 0 & 2 \\
0 & 0 & 0 & 6 & 5 & 0 & 1 & 0 & 0 & 4 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 208 factors with signed exponents of absolute value up to 70760531059040; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 0 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 4428y^3 + (52s - 26)\, y^2 - 4028269y + (-7412s + 3706)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 8856*y^8 + 11550646*y^6 + 8818708996*y^4 + 3543006704241*y^2
+ 903974962796860
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{9195119}{5023060313370901} + (-\frac{17324}{5023060313370901})\, s, \qquad J = \begin{pmatrix} 1381 & 823 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1381$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $67209581260800 = 2^{12} \cdot 3^4 \cdot 5^2 \cdot 11^2 \cdot 167 \cdot 401$:

$$\begin{pmatrix} 530378640 & 17582880 & 51540192 & 488681036 & 318011218 & 144283882 & 462488274 & 432881326 & 361755195 & 431406302 \\ 0 & 40 & 0 & 0 & 24 & 0 & 19 & 11 & 38 & 27 \\ 0 & 0 & 396 & 2 & 394 & 152 & 116 & 175 & 60 & 68 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 205 factors with signed exponents of absolute value up to 1189098449411209; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 499y^3 + (-20s + 10)\, y^2 + 518780y + (9630s - 4815)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 998*y^8 + 1286561*y^6 + 7099555940*y^4 - 6069153712100*y^2 +
1525942450920375
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2727091}{7515177189376} + \left(\frac{69}{7515177189376}\right) s, \qquad J = \begin{pmatrix} 376 & 102 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 376$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $62815232425905 = 3^2 \cdot 5 \cdot 29 \cdot 283 \cdot 170085787$:

$$\begin{pmatrix} 20938410808635 & 11959530549276 & 1777670601763 & 16445998551337 & 12824007308869 & 16347671763141 & 4643088919581 & 10145134163698 & 1355123928888 & 7583120609209 \\ 0 & 3 & 0 & 0 & 1 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 423 factors with signed exponents of absolute value up to 47343022488470165008569184540; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 499y^3 + (-20s + 10)\, y^2 + 518780y + (9630s - 4815)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 998*y^8 + 1286561*y^6 + 7099555940*y^4 - 6069153712100*y^2 +
1525942450920375
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{423739183}{478923089887538176} + (-\frac{134871}{478923089887538176})\, s, \qquad J = \begin{pmatrix} 3436 & 2514 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3436$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $62815232425905 = 3^2 \cdot 5 \cdot 29 \cdot 283 \cdot 170085787$:

$$\begin{pmatrix} 20938410808635 & 11959530549276 & 1777670601763 & 16445998551337 & 12824007308869 & 16347671763141 & 4643088919581 & 10145134163698 & 1355123928888 & 7583120609209 \\ 0 & 3 & 0 & 0 & 1 & 1 & 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 418 factors with signed exponents of absolute value up to 51636876843506689906245173985; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 499y^3 + (-20s + 10)\,y^2 + 518780y + (9630s - 4815)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 998*y^8 + 1286561*y^6 + 7099555940*y^4 - 6069153712100*y^2 +
1525942450920375
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{9195119}{5023060313370901} + (-\frac{17324}{5023060313370901})\,s, \qquad J = \begin{pmatrix} 1381 & 823 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1381$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $25097312214875 = 5^3 \cdot 16943 \cdot 11850233$:

$$\begin{pmatrix}
1003892488595 & 977215999005 & 476422651313 & 818852914870 & 310751809317 & 717630864284 & 617464530756 & 25036455751 & 271350166264 & 328971762283 \\
0 & 5 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 4 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 5 & 4 & 0 & 0 & 0 & 0 & 3 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 419 factors with signed exponents of absolute value up to 569271461831460500179794063335; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a+b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1408y^3 + (2s-1)\,y^2 + 700226y + (2118s - 1059)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2816*y^8 + 3382916*y^6 + 2037654551*y^4 + 629719261006*y^2 +
73813787857935
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2727091}{7515177189376} + \left(\frac{69}{7515177189376}\right) s, \qquad J = \begin{pmatrix} 376 & 102 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 376$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $414256155720235 = 5 \cdot 47 \cdot 4523 \cdot 389739587$:

$$\begin{pmatrix} 414256155720235 & 216419549108434 & 43138126896018 & 231518822387282 & 32584464217621 & 186269003539725 & 344622710039400 & 44631480178161 & 284266807696088 & 48357685988998 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 617 factors with signed exponents of absolute value up to 14337465750007525015251837496378193492201913120583600376342498; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (1 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a+b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1408y^3 + (2s - 1)\, y^2 + 700226y + (2118s - 1059)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2816*y^8 + 3382916*y^6 + 2037654551*y^4 + 629719261006*y^2 +
73813787857935
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{423739183}{478923089887538176} + (-\frac{134871}{478923089887538176})\, s, \qquad J = \begin{pmatrix} 3436 & 2514 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3436$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $69306691386400 = 2^5 \cdot 5^2 \cdot 13^2 \cdot 17 \cdot 30154321$:

$$\begin{pmatrix}
66641049410 & 56978271200 & 49062074452 & 59342101014 & 16581128900 & 14481436396 & 33317123900 & 44019062526 & 7830719769 & 20504256314 \\
0 & 10 & 6 & 2 & 4 & 4 & 2 & 7 & 8 & 0 \\
0 & 0 & 26 & 24 & 14 & 21 & 23 & 22 & 14 & 25 \\
0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 625 factors with signed exponents of absolute value up to 7792410538702590794282180684740183341891447753266426908361176; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 1408y^3 + (2s - 1)\, y^2 + 700226y + (2118s - 1059)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 2816*y^8 + 3382916*y^6 + 2037654551*y^4 + 629719261006*y^2 +
73813787857935
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{9195119}{5023060313370901} + (-\frac{17324}{5023060313370901})\, s, \qquad J = \begin{pmatrix} 1381 & 823 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1381$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9408299058511 = 17 \cdot 47^2 \cdot 250533887$:

$$\begin{pmatrix} 200176575713 & 83728177730 & 82961865025 & 166916732692 & 54008305564 & 153337427120 & 10631634609 & 164332355001 & 128498306374 & 178559875355 \\ 0 & 47 & 8 & 46 & 0 & 20 & 0 & 30 & 0 & 4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 617 factors with signed exponents of absolute value up to 14128213608025858302661376278261485375132975994065079540896671; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (4 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 459)\, y^3 + (-32s + 94026)\, y^2 + (20s + 2384885)\, y + (2920s + 4198015)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 918*y^8 + 188939*y^7 + 21247444*y^6 + 970324130*y^5 +
22829023855*y^4 + 327392360075*y^3 + 3408867996825*y^2 + 2195250340
2250*y + 157933526841625
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2727091}{7515177189376} + (\frac{69}{7515177189376})\, s, \qquad J = \begin{pmatrix} 376 & 102 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 376$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $34337724625 = 5^3 \cdot 4513 \cdot 60869$:

$$\begin{pmatrix} 1373508985 & 1201641340 & 607693915 & 670993078 & 407497101 & 201241587 & 521146528 & 940738925 & 998955740 & 1007758079 \\ 0 & 5 & 0 & 0 & 1 & 1 & 0 & 0 & 3 & 0 \\ 0 & 0 & 5 & 0 & 0 & 4 & 4 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 566 factors with signed exponents of absolute value up to 10315579000624460349595; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 459)\, y^3 + (-32s + 94026)\, y^2 + (20s + 2384885)\, y + (2920s + 4198015)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 918*y^8 + 188939*y^7 + 21247444*y^6 + 970324130*y^5 +
22829023855*y^4 + 327392360075*y^3 + 3408867996825*y^2 + 2195250340
2250*y + 157933526841625
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{423739183}{478923089887538176} + (-\frac{134871}{478923089887538176})\, s, \qquad J = \begin{pmatrix} 3436 & 2514 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3436$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $44777455405 = 5 \cdot 17 \cdot 526793593$:

$$\begin{pmatrix} 44777455405 & 27194151013 & 5550849355 & 34580100438 & 15093253168 & 21803145289 & 32529052248 & 24294814835 & 31310347811 & 15977002534 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 596 factors with signed exponents of absolute value up to 205188566567360227427l7; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 459)\, y^3 + (-32s + 94026)\, y^2 + (20s + 2384885)\, y + (2920s + 4198015)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 918*y^8 + 188939*y^7 + 21247444*y^6 + 970324130*y^5 +
22829023855*y^4 + 327392360075*y^3 + 3408867996825*y^2 + 2195250340
2250*y + 157933526841625
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{9195119}{5023060313370901} + \left(-\frac{17324}{5023060313370901}\right) s, \qquad J = \begin{pmatrix} 1381 & 823 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1381$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $159260256386111 = 159260256386111$:

$$\begin{pmatrix} 159260256386111 & 60742635824748 & 2085117131728 & 40112618317426 & 118164677014251 & 124257340947545 & 99753669023153 & 70306056152814 & 41757426177213 & 76508174887371 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 595 factors with signed exponents of absolute value up to 928586893014760384270l8; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3691y^3 + (18s - 9)\,y^2 + 183575y + (-550s + 275)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 7382*y^8 + 13990631*y^6 + 3976118285*y^4 - 292099987625*y^2 +
4977496459375
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 2 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{2727091}{7515177189376} + (\frac{69}{7515177189376})\,s, \qquad J = \begin{pmatrix} 376 & 102 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 376$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $41287842623375 = 5^3 \cdot 1307 \cdot 2179 \cdot 115979$:

$$\begin{pmatrix} 1651513704935 & 197082804290 & 681018395530 & 575917572615 & 522103902825 & 860373300322 & 1146319549058 & 323950215745 & 1334181509219 & 655367054469 \\ 0 & 5 & 0 & 1 & 0 & 0 & 2 & 2 & 0 & 1 \\ 0 & 0 & 5 & 1 & 4 & 4 & 3 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 471 factors with signed exponents of absolute value up to 265325962754132; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3691y^3 + (18s - 9)\, y^2 + 183575y + (-550s + 275)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 7382*y^8 + 13990631*y^6 + 3976118285*y^4 - 292099987625*y^2 +
4977496459375
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 2 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{423739183}{478923089887538176} + (-\frac{134871}{478923089887538176})\, s, \qquad J = \begin{pmatrix} 3436 & 2514 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3436$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $175224734029175 = 5^2 \cdot 271 \cdot 44563 \cdot 580379$:

$$\begin{pmatrix}
35044946805835 & 34948528340994 & 31952374147505 & 20022178114138 & 31720664882343 & 5856137943217 & 28804890454724 & 3076278950829 & 24578025479534 & 1991843914734 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 5 & 0 & 0 & 4 & 1 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 469 factors with signed exponents of absolute value up to 21007029185640; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3691y^3 + (18s - 9)\, y^2 + 183575y + (-550s + 275)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 7382*y^8 + 13990631*y^6 + 3976118285*y^4 - 292099987625*y^2 +
4977496459375
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 2 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{9195119}{5023060313370901} + (-\frac{17324}{5023060313370901})\, s, \qquad J = \begin{pmatrix} 1381 & 823 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 1381$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $175224734029175 = 5^2 \cdot 271 \cdot 44563 \cdot 580379$:

$$\begin{pmatrix} 35044946805835 & 34948528340994 & 31952374147505 & 20022178114138 & 31720664882343 & 5856137943217 & 28804890454724 & 3076278950829 & 24578025479534 & 1991843914734 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 4 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 466 factors with signed exponents of absolute value up to 194953005306918; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 3 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.